

Estimation of GPS accuracy of Maxi-Phyling V2.3

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Abstract: In the realm of sports science, the utilization of GPS localization is widespread for the acquisition of data related to location, distance and speed. The aim of this study is to assess GPS accuracy of the latest GPS chip incorporated in the Maxi-Phyling V2.3. Both static and dynamic tasks were analyzed to estimate accuracy of the GPS measurements. The obtained results indicate errors in distance and speed estimations that are under 1 meter and 1 km/h, respectively. In summary, the Maxi-Phyling emerges as a reliable and consistent GPS unit suitable for the precise measurement of location, distance and speed.

1. Introduction

The Global Positioning System (GPS) is a satellite-based radio positioning system, functioning through communication between a GPS chip and satellites orbiting the Earth. Each satellite consistently transmits its precise position and transmission time provided by its atomic clock.

GPS units on Earth gather diverse information from these orbiting satellites. Positional data, including latitude, longitude and altitude, are determined through triangulation of satellites positions [1]. A minimum of four satellites is required for position estimation using this method. Simultaneously, the GPS chip records the time from each satellite. Parameters such as Position Dilution of Precision (PDOP) and the number of satellites serve as important indicators of GPS accuracy [2]. PDOP is a parameter gauging the robustness of the current satellites configuration or geometry, with a PDOP close to 1 or below indicating excellent positioning accuracy.

The parameters of location, distance and speed are commonly employed in sports sciences to track athlete performances and monitor training sessions [3]. As a result, precise GPS data is imperative for

the athlete's support staff to enhance performance analysis.

This study aims to assess the accuracy of the Maxi-Phyling unit concerning location through a static analysis, and speed and distance measurements through a dynamic study.

2. Methods

The Maxi-Phyling (Figure 1) used in this study is an acquisition unit with integrated sensors such as IMU and GPS. The Maxi-Phyling V2.3 is equipped with the M10Q GPS chip.



Figure 1 : Maxi-Phyling unit

The assessment of the GPS unit accuracy will be conducted through two distinct studies: a static study and a dynamic study.

Static study

To precisely assess GPS accuracy of the Maxi-Phyling, the unit is positioned at a fixed location with known coordinates (obtained from Google Maps). Data are recorded over a period of 20 minutes at a frequency of 10 Hz.

Accuracy, variability, number of satellites and PDOP are variables of interest. Accuracy is defined as the distance between the known location and the mean location recorded by the Maxi-Phyling. Variability is calculated as the average distance between the mean location and each GPS location recorded by the Maxi-Phyling.

Dynamic study

To assess distance and speed accuracy and repeatability during dynamic activities, the Maxi-Phyling is fixed on a bike and a cyclist completes eight laps along a 400-meter athletics track. The duration of each lap is recorded with a stopwatch to calculate the reference mean speed.

3. Results

3.1 Static results

Accuracy and variability, mean number of satellites and mean PDOP are shown on Table 1.

Table 1 : Mean results of GPS static accuracy

	Accuracy (m)	Variability (m)	Number of satellites	PDOP
Maxi-Phyling	1.3	0.58	25.85	1.01

3.2 Dynamic results

Mean errors of distances and speeds measured by the Maxi-Phyling unit are presented in Table 2.

Table 2 : Mean errors of distance and speed estimations

	Distance error (m)	Speed error (km/h)
Mean values ($\pm$ SD)	0.59 ± 0.34	0.27 ± 0.18

4. Discussion

The aim of this study was to assess the accuracy of the Maxi-Phyling unit concerning location, distance, and speed measurements.

The results of the static study reveal that the positional accuracy of the Maxi-Phyling unit is approximately 1.3 meters in comparison to a known location. This determination is based on the mean position measured by the Maxi-Phyling, excluding the first minute of recording. Indeed, as shown on Figure 2, when the GPS chip is turned on, it needs some time to acquire satellites and reach its optimal operating mode.

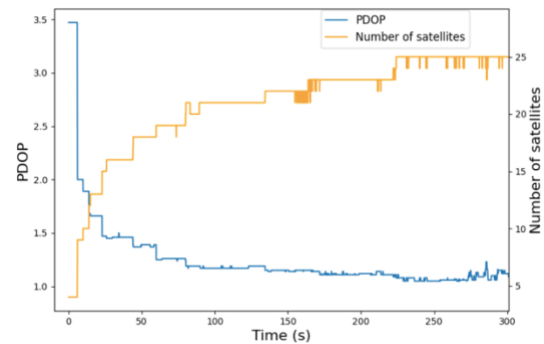


Figure 2 : Evolution of the number of satellites and PDOP during the first 5 min of record.

The mean variability of the measured position (0.58m) indicates low fluctuations in location. This stability is favorable for applications requiring consistent and reliable positional data.

Moreover, the mean PDOP close to 1 and the average number of satellites (25.85) suggest a robust satellite configuration during the data collection period. A PDOP near 1 is indicative of excellent positioning precision, further supporting the reliability of the measurements obtained from the Maxi-Phyling unit.

In the dynamic study, the obtained results indicate average errors in distance and speed estimations of 0.59 m and 0.27 km/h respectively. Additionally, the observed low standard deviation values indicate a high degree of reliability in the results. These outcome demonstrate a high level of accuracy for both distance and speed estimations during dynamic activities.

5. Conclusion

The incorporation of the latest GPS chip in the Maxi-Phyling V2.3 has proven to provide accurate and repeatable measurements of location, distance and speed. The device demonstrates

repeatability and precision, making it suitable for applications in sports science or any other context where accurate and consistent GPS data is essential.

6. Bibliography

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